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Intrinsic Peroxidase-Mimicking Ir Nanoplates for Nanozymatic Anticancer and Antibacterial Treatment

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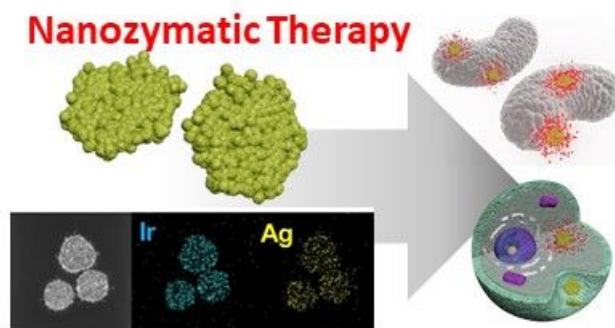
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KEYWORDS. antibacterial, anticancer, iridium nanoplates, nanozyme, peroxidase

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ABSTRACT

The study of inorganic nanozymes to overcome the disadvantages of bio-enzymes such as the requirement of optimized reaction conditions and lack of durability against environmental factors is one of the most significant research topics at present. In this work, we comprehensively analyzed the intrinsic peroxidase-like activity of Ir-based nanoparticles, the biological and nanozymatic potentials of which have not yet been explored. These particles were synthesized by the galvanic replacement of Ag nanoplates with Ir. Through the confirmed peroxidase-like activity and hydrogen peroxide decomposition with free radical generation facilitated by these particles, the antibacterial and anticancer effects were successfully verified *in vitro*. The nanozyme-based therapeutic effect observed at concentrations at which these nanoparticles do not show cytotoxicity suggests that it is possible to achieve more precise and selective local treatment with these particles. The observed highly efficient peroxidase-like activity of these nanoparticles is attributed to the partially mixed composition of Ir-Ag-IrO₂ formed through the galvanic replacement reaction in the synthetic process.

INTRODUCTION

In recent years, nanoparticles with controlled size, morphology, and composition have been extensively applied in various fields owing to their unique physicochemical properties such as localized surface plasmon resonance (LSPR),¹⁻³ magnetic susceptibility,⁴⁻⁶ and a high surface-to-volume ratio.^{7,8} In particular, photo-triggered responses, including photothermal conversion^{9,10} and photocatalytic activity,^{11,12} which are conventionally observed with plasmonic Au or Ag nanoparticles have been of great advantage for application in biological and catalytic applications. For example, drug delivery systems based on their structural aspects, such as the high surface-to-volume ratio and several high-complexity structural features, have been developed for bio-applications, and there have been many reports on the study of nanoparticles with innate biologically active compositions based on metals such as Ag and Rh.¹³⁻¹⁵ In addition, high-energy heat sources derived from photothermal conversion and reactive oxygen species (ROS) generated by photodynamic treatment *via* the laser irradiation of designated regions are used in combination with chemotherapy to eliminate several lesions, including cancer.^{16,17} However, such phototherapeutic approaches have certain limitations such as the unintended damage of the adjacent tissues by thermal radiation in photothermal therapy (PTT) or a rapid decay of the efficiency owing to the depletion of dissolved oxygen species during photodynamic therapy (PDT), which should be overcome.

For the control of biosystems and *in vitro* implementation, nanozymes (bio-enzyme mimetic inorganic materials with a compound word derived from “nanoparticle” and “enzyme”) have received substantial attention. Since nanozymes are basically composed of inorganic elements, they provide the following advantages: ease of handling, highly persistent catalytic activity, and broad pH/temperature tolerance compared to bio-enzymes.^{18,19} Heretofore, numerous nanozymes

with metal or metal oxide compositions that mimic bio-enzymes have been reported including those showing peroxidase-, oxidase-, catalase-, and superoxide dismutase-like activities.²⁰⁻²² Among these, the peroxidase-like activity of nanozymes, which is directly related to ROS generation for synergistic cancer therapy, has been studied actively. Even complex applications based on the interaction of multi-nanozymatic reactions from single nanoparticles beyond the mono-catalytic reactions are being developed. Following the same trend, the development of nanozymes based on advanced materials such as doped carbon nanomaterials and Ru–Te alloys has continued, and the design of nanoparticles with unprecedented structures and compositions is critical for the advancement of this field.²³ Based on their ROS generation characteristics, developed nanozymes are considered as promising candidates for medical applications including antibacterial disinfection and cancer therapy.²⁴⁻²⁹

Herein, we propose iridium nanoplates (IrNPs) with rough surface morphology as nanozymes. They show antibacterial and anticancer effects through the intrinsic peroxidase-like activity based on their elemental composition (**Figure 1**). IrNPs synthesized by hydrothermal galvanic replacement based on sacrificial Ag nanoplates show Ir–Ag alloy composition and have a uniform size of about 90 nm with excellent colloidal stability.³⁰ Moreover, the prepared IrNPs exhibit effective peroxidase-like nanozymatic activity and show Michaelis-Menten kinetics for various substrates. From the established biocompatible concentrations of IrNPs and hydrogen peroxide (H₂O₂) treatment, we confirmed that they induce cancer cell apoptosis through ROS generation *in vitro*. Further, their nanozymatic antibacterial effect against *E. coli* was also found to be effective. It is confirmed that effective peroxide decomposition is induced through the Ir(III)/Ir(IV) multivalent redox reactions, and this study is expected to serve as a guide for the development of therapeutic nanozymes based on platinum group transition metals.

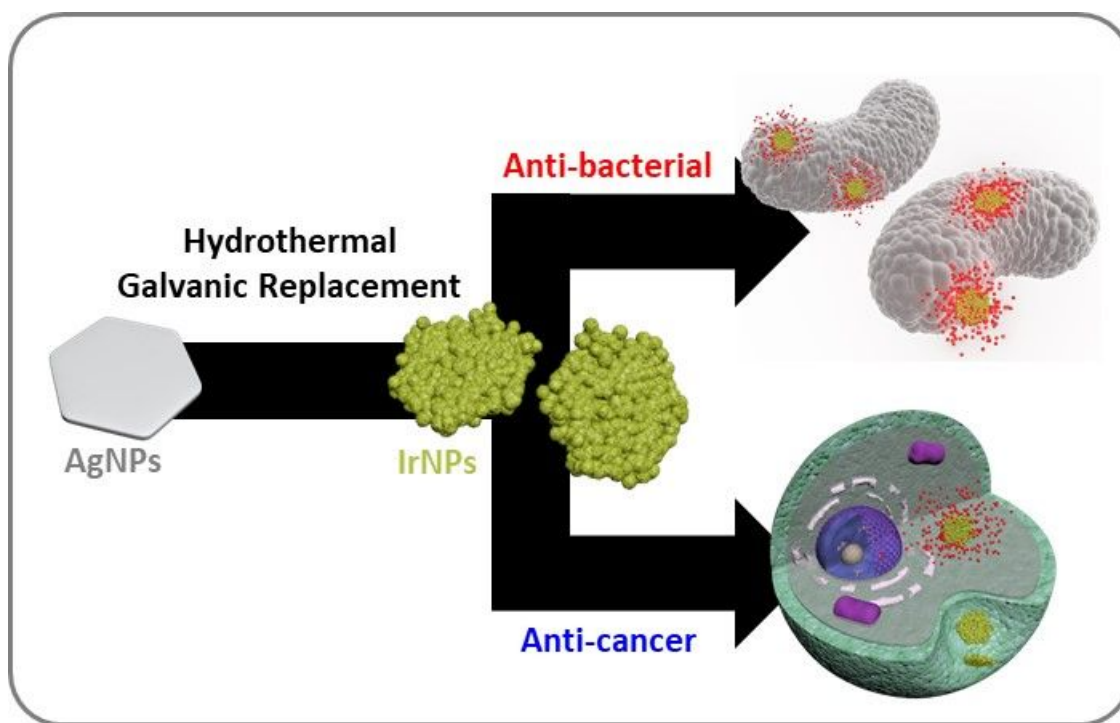


Figure 1. Schematic illustration of the anti-bacterial and anti-cancer potential of nanozymatic IrNPs.

RESULTS AND DISCUSSION

Synthesis and characterization of IrNPs

IrNPs were synthesized by a previously reported hydrothermal galvanic replacement method using silver nanoplates as sacrificial templates (**Figure S1**). Briefly, the replacement is driven by a spontaneous reaction owing to the higher standard reduction potential of iridium ($\text{Ir}^{3+}|\text{Ir} = 1.15 \text{ V}$ vs. standard hydrogen electrode (SHE)) compared to that of silver ($\text{Ag}^{+}|\text{Ag} = 0.7996 \text{ V}$ vs. SHE).³¹ However, the replacement did not proceed in ambient conditions, possibly due to the unknown activation energy barrier to nanostructure formation through the reaction between Ir and Ag. To overcome this barrier, the hydrothermal galvanic replacement was performed at a high temperature

of 200 °C to achieve a spontaneous replacement reaction and subsequent secondary growth of IrNPs owing to the high-energy environment. The so-prepared IrNPs were re-dispersed in deionized (DI) water for subsequent characterizations and applications.

The transmission electron microscopy (TEM) image of IrNPs shows particles with rough surface morphology and a transverse diameter of ~90 nm (**Figure 2a**). Owing to the galvanic replacement of Ag by Ir, the color of the Ag nanoplate dispersion changed from vivid blue to black (**Figure 2b**; inset). This change in the optical characteristic was additionally confirmed by UV-vis spectroscopy. The UV-vis spectra in **Figure 2b** show that the distinctive LSPR peaks of the Ag nanotemplate (338 nm and 765 nm for out-of-plane quadrupole and in-plane dipole resonances, respectively) totally diminished, while the LSPR peak of the Ir particles in the UV region appeared, following the reaction. The colloidal dispersity of IrNPs was verified by dynamic light scattering and ζ -potential measurements. The identified highly negative ζ -potential of IrNPs (-25.4 ± 0.6 mV) in $1\times$ phosphate buffered saline (PBS) confirmed the excellent colloidal stability. Further, a hydrodynamic diameter of 98.3 ± 2.2 nm with a poly dispersity index of 0.132 ± 0.009 was determined, which is in the same range as that found by TEM studies, despite the platelet morphology (**Figure 2c**).

The structure and composition of IrNPs, including the crystallinity and elemental oxidation states, were further verified through X-ray techniques and high-angle annular dark field scanning TEM (HAADF-STEM). According to the deconvoluted Ir 4f and Ag 3d X-ray photoelectron spectra (XPS), IrNPs are composed of metallic Ir (Ir^0 : 60.8 and 63.8 eV for $4f_{7/2}$ and $4f_{5/2}$, respectively) and Ag (Ag^0 : 367.3 and 373.3 eV for $3d_{5/2}$ and $3d_{3/2}$, respectively), and a small quantity of IrO_2 (Ir^{4+} : 61.6 and 64.8 eV for $4f_{7/2}$ and $4f_{5/2}$, respectively) due to surface oxidation (**Figure 2d** and **2e**).^{32,33} The X-ray diffraction (XRD) pattern of the IrNP film on Si wafer also indicated the alloyed

composition. The XRD pattern clearly suggests the co-existence of metallic Ir (JCPDS 06-0598) and Ag (JCPDS 65-2871) along with surface-oxidized IrO₂ (JCPDS 15-870) (**Figure 2f**).³⁴⁻³⁶ High-resolution (HR)-TEM and HAADF-STEM/energy dispersive spectral (EDS) mapping provided additional information on the IrNPs. The HR-TEM and fast Fourier transform (FFT) image clearly indicated the face centered cubic (*fcc*) crystalline structure of IrNPs with d-spacing of 0.224 nm from Ir(111) lattice structure (**Figure S2**). The HAADF-STEM/EDS mapping revealed the Ir–Ag alloy composition of IrNPs (**Figure 2g** and **S3**). The quantitative Ir:Ag ratio was determined by inductively coupled plasma mass spectrometry, which revealed a higher content of Ir (2.001 μg Ir/mL) than that of Ag (0.324 μg Ag/mL). It was confirmed that some Ag is derived from alloy formation during the galvanic replacement and the reduction of dissolved Ag cations through secondary growth. Remarkably, the Ag remaining in the particles was stable even upon etching with H₂O₂.

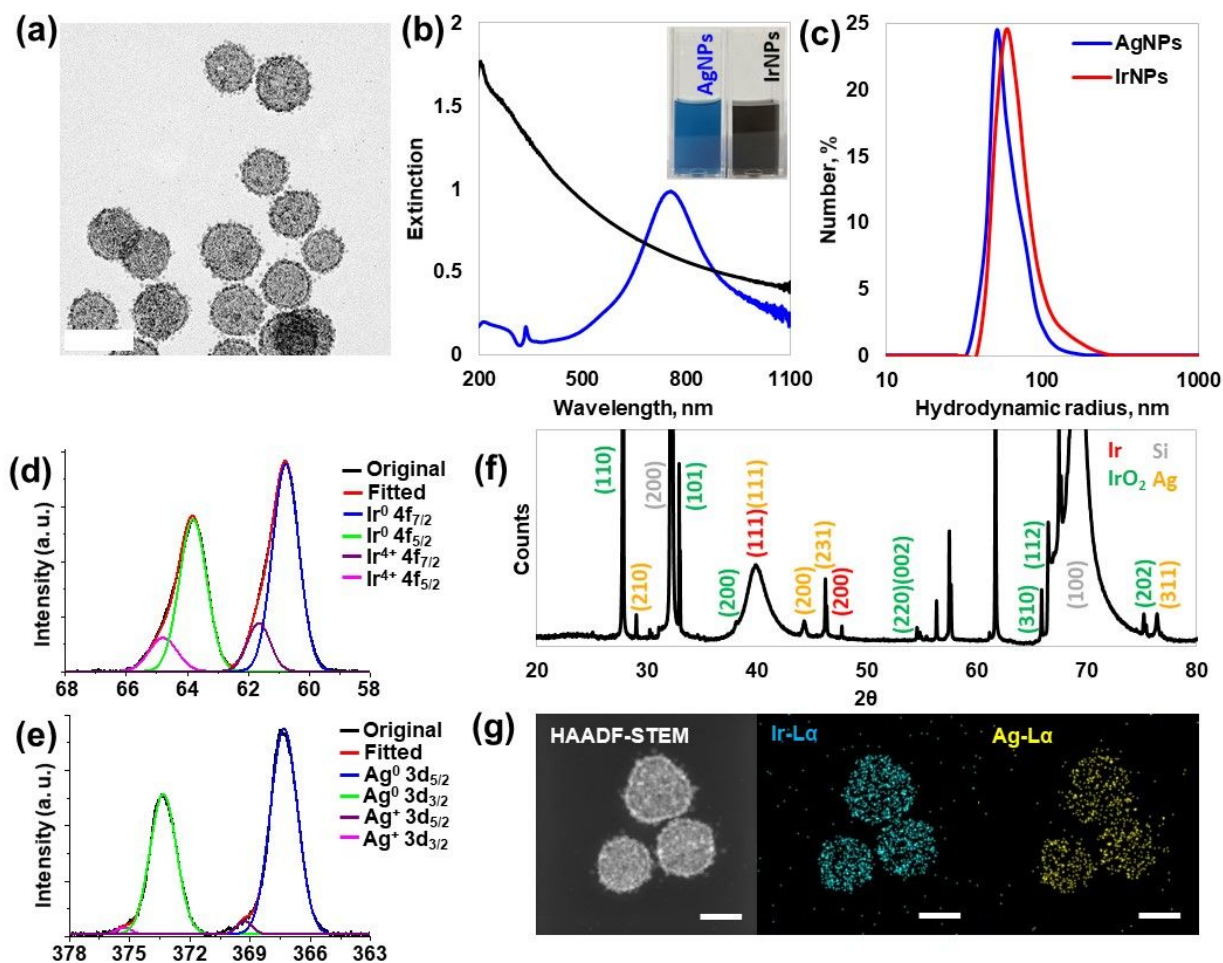


Figure 2. Characterization of IrNPs. (a) TEM image showing monodisperse platelets of IrNPs with a transverse diameter of ~90 nm (scale bar, 100 nm). (b) UV-vis spectra and digital photographs (inset) of the AgNP template and IrNPs. (c) DLS data of sacrificial AgNPs and IrNPs in 1× PBS. Deconvoluted XPS spectra of (d) Ir 4f and (e) Ag 3d regions of the IrNPs. (f) XRD pattern and (g) HAADF-STEM/EDS mapping images (scale bar, 50 nm) of IrNPs.

Peroxidase-like activity of IrNPs

To evaluate the potential of the IrNP as a nanozyme, model indicators including 3,3',5,5'-tetramethylbenzidine (TMB),^{37,38} o-phenylenediamine (OPDA),^{39,40} and dopamine hydrochloride (DOPA) were used in a H₂O₂-mediated reaction in a cuvette system for verifying its peroxidase-like activity.⁴¹ Reaction with the hydroxy radicals (HO·) generated by H₂O₂ decomposition at the surface of the peroxidase-like nanozyme is the common mechanism by which the indicator molecule is oxidized, which results in changes in its color and absorption spectrum. Before the addition of the substrate, H₂O₂, none of the three indicators showed absorption in the visible-near infrared (vis-NIR) region in a pH 5 environment; however, a distinctive peak started to appear in each case, at 632 nm for TMB, 450 nm for OPDA, and 490 nm for dopamine, as the nanozymatic reaction progressed (**Figure 3a-c**). As a result of the increase in the absorbance, the color of the reaction solution also changed from transparent to blue (TMB), yellow (OPDA), or pink (dopamine) (**Figure 3d**). As all three indicators exhibited the same characteristics, to further confirm the nanozymatic activity, detailed analysis was performed against TMB, which shows the highest efficiency in the colorimetric assay with a rapid increase in its absorbance. As expected, it was confirmed that the observed peroxidase activity is highly dependent on the concentration of IrNPs, which is the main constituent in the reaction system that shows the nanozymatic property (**Figure S4**).

To confirm the tendency of the reaction rate according to the H₂O₂ concentration, the absorbance at 653 nm was plotted as a two-parameter hyperbola regression against the average rate for 10 s from the reaction. The reaction rate clearly followed the Michaelis-Menten kinetics, from which the maximum velocity ($V_{\max} = 67.94 \pm 3.83 \mu\text{M}\cdot\text{s}^{-1}$) and Michaelis constant ($K_M = 0.3193 \pm 0.0517 \text{ M}$) were calculated (**Figure 3e and S5**). The same tendency was observed when the concentration

of TMB serving as a substrate as well as indicator was varied ($V_{\max} = 38.95 \pm 1.16 \mu\text{M}\cdot\text{s}^{-1}$ and $K_M = 131.7368 \pm 11.7658 \mu\text{M}$) (**Figure 3f** and **S6**). Further, the peroxidase-like activity of IrNPs showed a different tendency with respect to pH. Through experiments based on standard buffer solutions with pH ranging from 1 to 13, the highest absorption intensity at 653 nm was identified at pH 4. When the pH was higher or lower than 4, a tendency of a gradual decrease in the absorption intensity was observed, and no activity was observed in highly acidic (pH 1) and basic (pH 8–13) conditions (**Figure 3g** and **S7**). The TMB color development according to the pH was additionally confirmed through digital imaging (**Figure 3g**; inset). It was observed that the higher the acidity is, the yellower the solution is, and not the vivid blue of oxidized TMB. Conventionally, the peroxidase-like activity of bio- and inorganic compounds induces the decomposition of H_2O_2 to produce the hydroxyl radical, which participates in the one-electron oxidation of TMB. The formation of the charge-transfer complex of the cationic radical of TMB (TMB_{ox} ; $\epsilon_{653 \text{ nm}} = 39,000 \text{ M}^{-1}\cdot\text{cm}^{-1}$) results in the widely known blue coloration. However, upon further reaction or acidification due to rapid enzymatic activity or a sufficiently acidic environment, a yellow TMB diimine compound (TMB_{di} ; $\epsilon_{450 \text{ nm}} = 59,000 \text{ M}^{-1}\cdot\text{cm}^{-1}$) is formed through two-electron oxidation (**Figure 3h**).^{42,43} To confirm the relative abundance of the two TMB species, the absorbance at the wavelength corresponding to each species was plotted. Considering the TMB_{ox} and acidified TMB_{di} species together, the highest peroxidase efficiency was found at pH 3 (**Figure 3i**). Based on the total oxidized TMB species, a peroxidase-like activity of more than ~70% was observed between pH 2 and 5, which can be effectively utilized in weakly acidic environments (pH 4.5–6.5) such as in late endosomes, lysozymes, and tumor tissues.

The peroxidase-like activity of IrNPs was observed only with a tendency to be difficult to predict for 180 s of the reaction at various temperature conditions (**Figure S8a**). The cationic radical

TMB_{ox} species exhibited the highest efficiency at 40 °C, like bio-enzymes, which resulted in the maximum activity at a specific temperature (**Figure S8b**). In addition, because IrNP is an inorganic nanozyme that is thermally stable, a relative activity of 79.5% of the maximum efficiency was calculated even at 80 °C, where biomolecules lose their activities. On the other hand, the concentration of TMB_{di}, which can also be considered as the final product of the intermediate TMB_{ox}, was observed to increase continuously with the increase in temperature (**Figure S8c**). This phenomenon suggests that the as-produced TMB_{ox} is consumed through a continuous reaction. Considering that variability and inaccuracy might originate from the average velocity based on the total reaction time (**Figure S9**; red), TMB_{ox} based on short-term velocity (10 s from the initiation) was plotted (**Figure S9**; blue). Thus, it was confirmed that the tendency of the initial velocity is the same as the temperature-dependent reaction rate increase of TMB_{di}. Based on this, it is inferred that additional consideration is required to measure the actual peroxidase activity, that is, several factors should be considered at the same time, instead of simply monitoring the absorbance of the cationic radical species.

In order to confirm the advantages in the effective working range of nanozymatic reaction and substrate binding affinity of the IrNPs, POD reaction was performed on a horseradish peroxidase (HRP), a conventional bioenzyme. As a result of POD reaction of HRP with TMB as substrate, HRP exhibited higher optimum pH range for enzymatic activity (pH 4 to 9; over 25% activity) than IrNPs (pH 2 to 6; over 25% activity). More drastic difference was revealed in the temperature-controlled test from 20 to 80 °C. Compare with the dramatic HRP activity decrease above 50 °C, IrNPs showed guaranteed nanozymatic activity in all tested temperatures (over 80%). From the substrate concentration test, higher K_M was calculated from the HRP (0.5696 ± 0.0694 M and $K_M = 201.1783 \pm 24.9798$ μ M for H₂O₂ and TMB, respectively) than IrNPs (0.3193 ± 0.0517 M and

131.7368 $\pm$ 11.7658 μ M for H₂O₂ and TMB, respectively), and it implied the lower binding affinity of HRP than IrNPs against all involved substrates.

A plausible mechanism of the peroxidase activity of IrNPs could be proposed from the redox potential of Ir species, the main component of IrNPs. From the previously described XPS characterization, it was confirmed that metallic Ir⁰ and its oxidized part Ir⁴⁺ coexist, and that redox reaction *via* Ir³⁺ is possible. The partial oxidation of Ir⁰ (Ir⁴⁺|Ir⁰ = 0.926 V, Ir⁴⁺|Ir³⁺ = 0.223 V, and Ir³⁺|Ir⁰ = 1.156 V vs. SHE) is sufficient to induce the peroxidase activity in acidic conditions in combination with the decomposition of H₂O₂ (H₂O₂, H⁺|H₂O = 1.776 V vs. SHE) and oxidation of TMB (TMB_{ox}|TMB_{red} = 1.13 V vs. SHE).^{44,45} The deconvoluted XPS spectrum of Ir 4f after the POD reaction exhibited the conversion of metallic Ir⁰ into oxidized and stable Ir⁴⁺ state in comparison with pristine IrNPs (**Figure S11**).

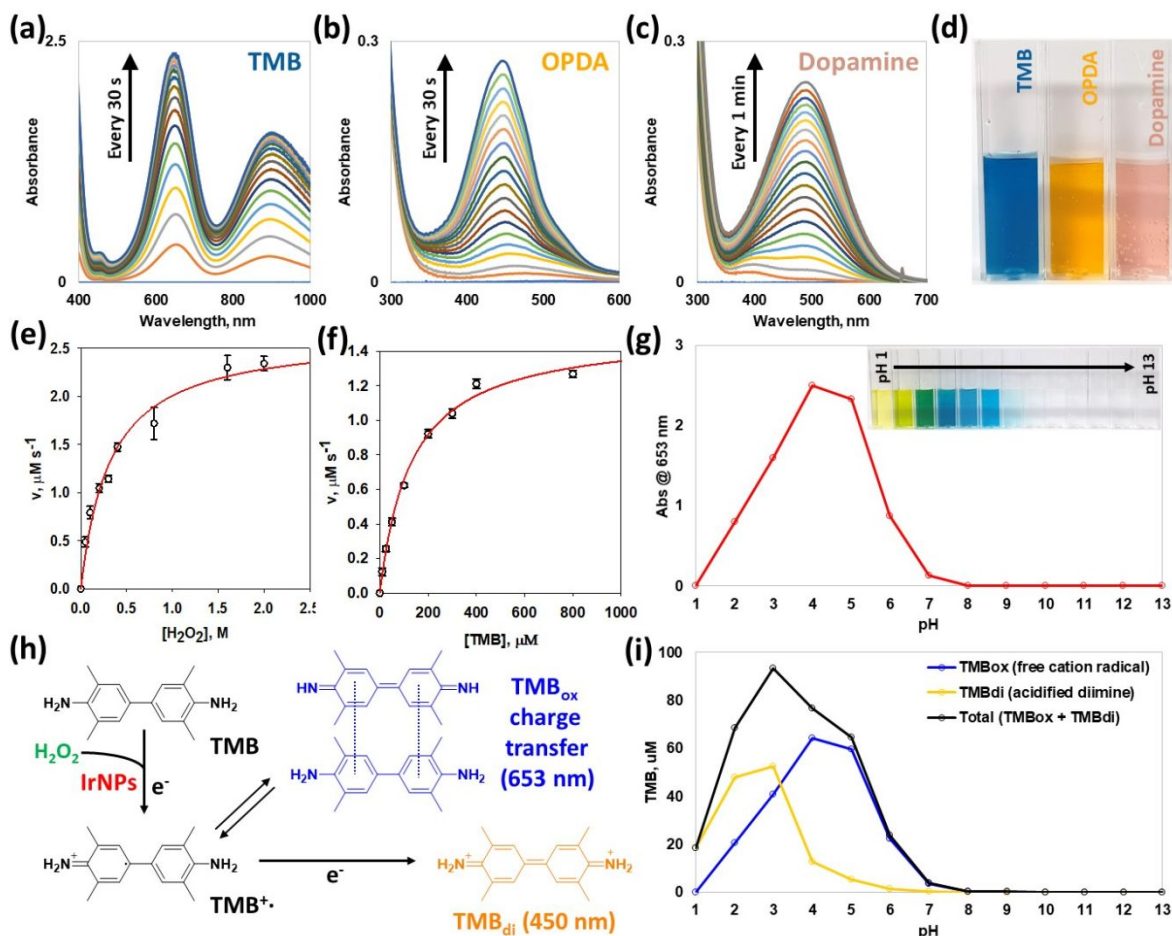


Figure 3. Peroxidase-like activity of IrNPs. UV-vis spectral monitoring of the peroxidase-like activity of IrNPs against model indicators including (a) TMB, (b) OPDA, and (c) dopamine, and their (d) digital photographs at the final time points. Peroxidase-like activity of IrNPs against different substrates: (e) H_2O_2 and (f) TMB. (g) pH dependent nanozymatic activity of IrNPs. Comparison of the peroxidase-like activities of IrNPs at different pH. (h) Types of TMB oxidation and color development. (i) pH-dependent peroxidase-like activity for one-electron oxidation (TMB_{ox}) and two-electron oxidation (TMB_{di}) processes.

Nanozymatic antibacterial effect of IrNPs

Although there have been several studies on the antibacterial application of organoiridium(III) complexes,^{46,47} the antibacterial effects of Ir-based nanomaterials are still unknown. From the recent discovery of the effective antibacterial effect of the nanoparticles of the same group 9 element, Rh, it has been speculated that the assessment of the potential of IrNPs for antibacterial applications in combination with their nanozymatic activity is extremely important in the biomedical field. To evaluate the synergistic antibacterial effect of IrNPs based on the intrinsic antibacterial effect of elemental Ir and that due to the nanozyme, we confirmed the effective concentration of H₂O₂ at which it did not show oxidative toxicity to *Escherichia coli* (*E. coli*). According to the optical density at 600 nm (OD₆₀₀),^{48,49} IrNPs exhibited intrinsic antibacterial effect in a dose-dependent manner (**Figure S12a**). In the case of H₂O₂ treatment, similar tendency of the decrease in the bacterial concentration was also confirmed (**Figure S12b**). To confirm that the antibacterial effect of the IrNPs is not due to the embedded Ag in the particle after the galvanic replacement reaction, IrNPs were incubated with 5% v/v H₂O₂ to potentially induce Ag etching and dissolution. According to the UV-vis spectra and TEM image, there was neither a spectroscopic shift nor structural transformation, respectively (**Figure S13**).

In order to confirm the antibacterial effect from direct reaction between the nanozyme mediator IrNPs and H₂O₂ as a substrate for ROS generation, the concentration of H₂O₂ was adjusted under a constant IrNP concentration (0.01 µg Ir/mL) and OD₆₀₀ was measured over time (**Figure 4a**). With the increase in the H₂O₂ concentration from 50 to 500 mM, more efficient antibacterial characteristics were clearly observed. The same tendency was identified when the amount of IrNPs was increased under a constant H₂O₂ concentration (200 µM) (**Figure 4b**). The OD₆₀₀ nm gradually increased because IrNPs absorbed at 600 nm (initial OD₆₀₀ 0 min = 0.026, 0.028, 0.054,

and 0.0025, 0.005, 0.01, and 0.02 $\mu\text{g Ir/mL}$, respectively); it was however deemed to be a reliable result from the difference between the initial absorbance and the OD600 value measured after 160 min ($\Delta\text{OD600}_{160\text{ min}} = 0.676, 0.422, 0.182, \text{ and } 0.094$ for 0.0025, 0.005, 0.01, and 0.02 $\mu\text{g Ir/mL}$, respectively). Finally, to demonstrate the intrinsic antibacterial effects of IrNPs and H_2O_2 and the synergistic nanozymatic antibacterial effect derived from dual treatment, the OD600 measurement was performed by setting 0.01 $\mu\text{g Ir/mL}$ and 200 μM as the optimized concentrations of IrNPs and H_2O_2 , respectively, suitable for unraveling the relative tendency (**Figure 4c**). Compared to the untreated control set ($\Delta\text{OD600}_{180\text{ min}} = 0.791$), mono-treated condition resulted in a slightly better antibacterial effect ($\Delta\text{OD600}_{180\text{ min}} = 0.770$ and 0.665, respectively, for H_2O_2 only and IrNPs only). Moreover, the dual treatment of IrNPs and H_2O_2 resulted in a markedly high efficiency ($\Delta\text{OD600}_{180\text{ min}} = 0.423$) with a delayed increase in the bacterial concentration until 120 min of incubation. Based on the OD600 of the control condition at 160 min, the synergistic factor of the combinatorial treatment of IrNPs and H_2O_2 was calculated to be 106.37, and thus the synergistic effect was confirmed clearly.^{50,51} According to the live/dead staining followed by fluorescent microscopic observation, nanozymatic antibacterial effect was clearly represented from the co-treatment of IrNPs and H_2O_2 against *E. coli*. Moreover, the SEM images showed bacterial membrane adsorption of IrNPs and membrane bursting (**Figure 4d-f**).

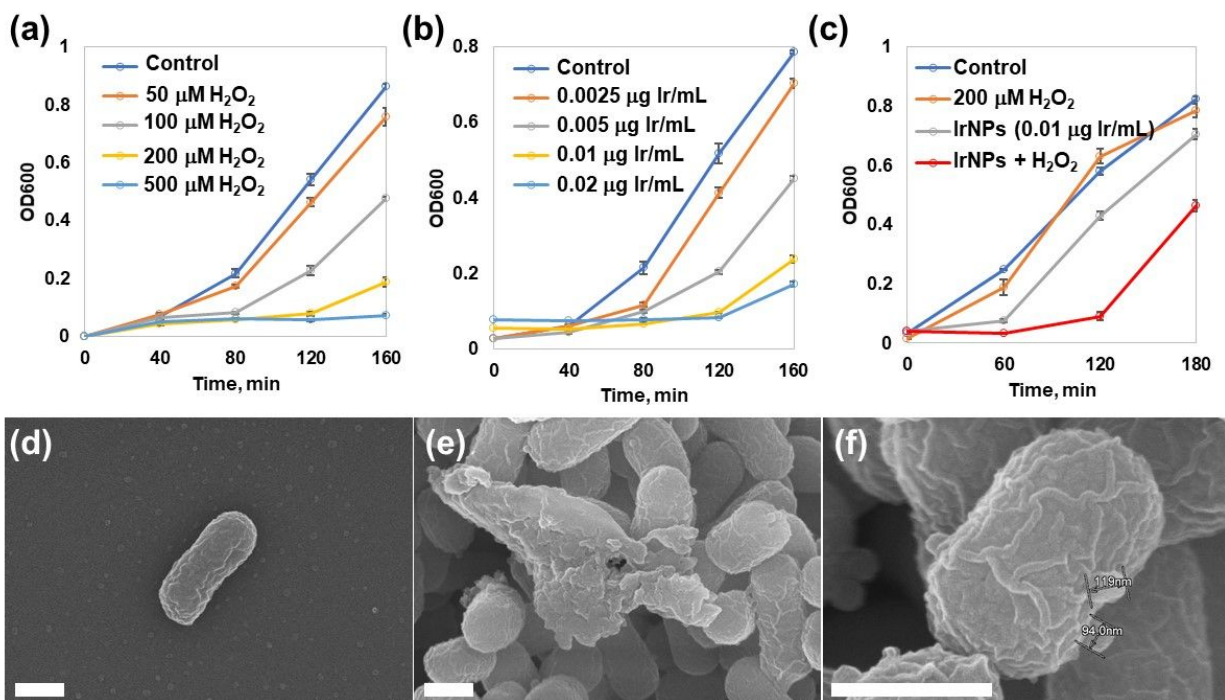


Figure 4. IrNP-based nanozymatic antibacterial effect against *E. coli*. Bacterial concentration was monitored from the OD at 600 nm against a (a) fixed IrNP concentration (0.01 $\mu\text{g Ir/mL}$) and variable H_2O_2 concentration, and (b) a fixed H_2O_2 concentration (200 μM) and variable IrNP concentration. (c) In comparison with the treatment of either H_2O_2 (orange) or IrNPs (grey), the dual-treatment condition (red) for the nanozymatic production of ROS led to synergistic antibacterial effect. (d-f) SEM image of *E. coli*. Compare to the (d) untreated control bacteria, IrNPs and H_2O_2 treatment induced (e) bursting and (f) surface adsorption on bacterial membrane. The scale bars are 500 nm.

In vitro nanozymatic anticancer characteristics of IrNPs

Before the anticancer therapeutic application, the intrinsic cytotoxicity of IrNPs was characterized *in vitro* against human cervical cancer cell line, HeLa. By employing 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) as a cell metabolic indicator, dose-dependent HeLa cell viability was successfully identified up to a concentration of 2.000 $\mu\text{g Ir/mL}$. According to the relative cell viability, no toxicity was observed at concentrations below 0.250 $\mu\text{g Ir/mL}$, and the cytotoxicity increased gradually beyond this concentration (**Figure 5a**). Based on these results, we chose 0.25 $\mu\text{g/mL}$ as the optimized concentration of IrNPs for investigating the nanozymatic response *in vitro* only, without the intrinsic particle toxicity.

Upon treatment with the optimized concentration of IrNPs (0.25 $\mu\text{g Ir/mL}$), cultured HeLa cells showed the same level of biocompatibility as that of untreated cells (**Figure 5b; control and NPs**). In addition, no specific cell death was observed even when only H_2O_2 was added to the cell culture media at concentrations ranging from 0.1 μM to 0.8 μM without the IrNPs (**Figure 5b; right columns**). However, when the cells were treated with IrNPs and incubated for a sufficient time to ensure internalization, a significant difference was identified when it was treated with H_2O_2 . In the process, although the IrNPs dispersed in the solution were removed through repeated washing and media replacement to prevent non-selective peroxidase activity, the H_2O_2 concentration-dependent cancer cell ablation was clearly observed (**Figure 5b (red columns)**; 94.0%, 76.6%, 59.4%, and 34.0% viability for 0.1, 0.2, 0.4, and 0.8 μM H_2O_2 addition). In order to confirm that the ROS generation by the peroxidase-like nanozymatic activity of IrNPs is the key mechanism, intracellular ROS staining with 5-(and-6)-carboxy-2',7'-dihydrofluorescein diacetate (carboxy- H_2DFFDA ; oxidative stress indicator) was further performed. Under the IrNP (0.25 $\mu\text{g Ir/mL}$) and H_2O_2 (0.8 μM) treatment, intense green fluorescence ($\lambda_{\text{ex}} = 492 \text{ nm} / \lambda_{\text{em}} = 520 \text{ nm}$) developed, as observed under a microscope (**Figure 5c**).

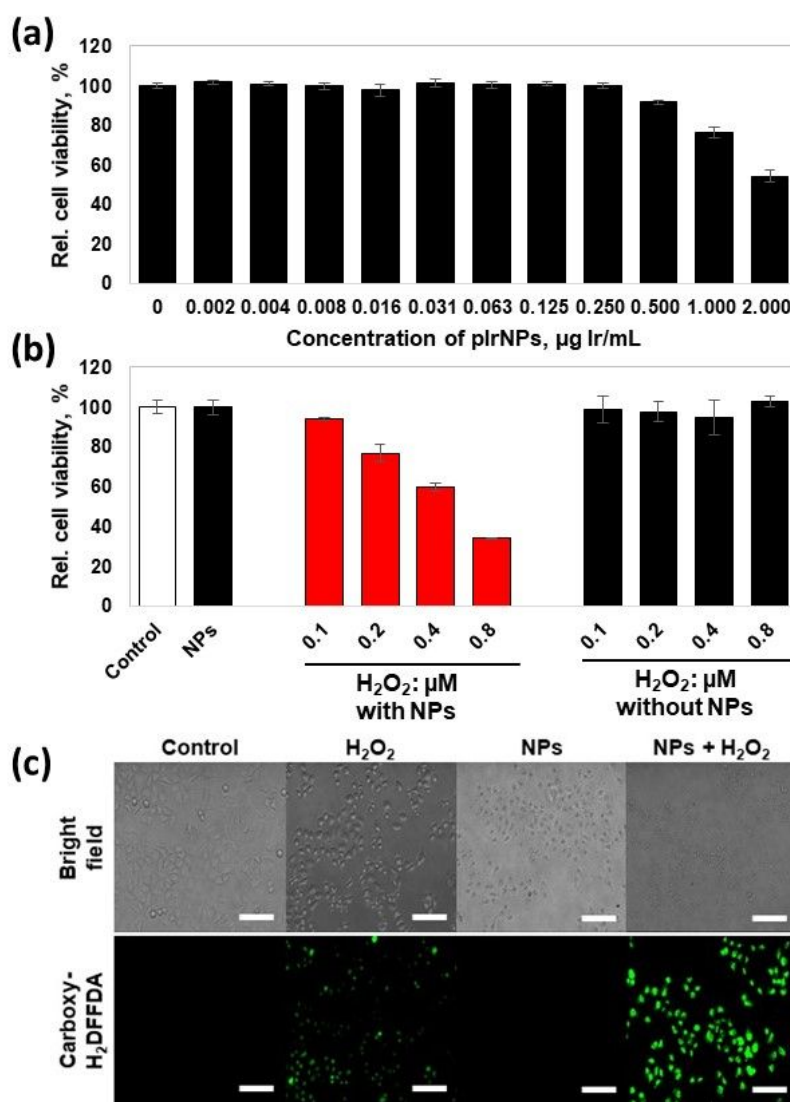


Figure 5. Peroxidase-like nanozymatic anticancer effect of IrNPs. (a) Cytotoxicity of IrNPs against HeLa cells. (b) Nanozymatic radical generation for cancer cell ablation. (c) Intracellular reactive oxygen species observed using the carboxy- H_2DFFDA indicator. Scale bar is 100 μm .

CONCLUSION

IrNPs with rough surface morphology were successfully synthesized through the hydrothermal galvanic replacement of Ag nanoplates. IrNPs composed of metallic Ir and Ag, as well as IrO₂ owing to surface oxidation, were shown to serve as nanozymes; they showed intrinsic peroxidase-like activity and triggered H₂O₂ decomposition to generate ROS with high efficiency. Consistent reaction rate trends under Michaelis-Menten kinetics were observed against the substrate concentration, reaction temperature, and IrNP amount. The ROS generated from the IrNP and H₂O₂ treatment at optimized concentrations, when they do not exhibit cytotoxicity, showed nanozymatic antibacterial and anticancer characteristics *in vitro*. Although nanozymes of various compositions and structures have been investigated actively and extensively, the verification of the biomedical potential of nanozymes based on group 9 elements remains to be explored. We believe that newer approaches and findings targeting Ir, a promising candidate for organic- and electrocatalyst field, will contribute to the expansion of the field toward the development of antibacterial and anticancer nanozymes.

EXPERIMENTAL

Materials. Silver nitrate, polyvinylpyrrolidone (Mw 29 kDa), trisodium citrate dihydrate, citric acid, L-ascorbic acid, sodium borohydride, iridium(III) chloride hydrate, dimethyl sulfoxide (DMSO), 3,3',5,5'-tetramethylbenzidine (TMB), o-phenylenediamine (OPDA), dopamine hydrochloride, horseradish peroxidase powder, Luria Bertani (LB) media, and 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) were purchased from Sigma-

Aldrich (St. Louis, MO, USA). Standard buffered solutions (pH 1 to 13) were purchased from Duksan (Gyeong-gi, Republic of Korea). Hydrogen peroxide was purchased from Samchun chemicals (Pyeongtaek, Republic of Korea). 10x phosphate buffered saline (PBS), Dulbecco's modified Eagle's medium (DMEM), fetal bovine serum (FBS), penicillin-streptomycin, and 0.05% trypsin-EDTA were purchased from WelGENE Inc. (Daegu, Republic of Korea).

All chemicals were used as received.

Synthesis of IrNPs

Preparation of sacrificial Ag nanoplates

For the preparation of Ag nanoseed, 0.152 g of polyvinylpyrrolidone (PVP, MW = 29 kDa) dissolved in 26.25 mL of deionized (DI) water, 250 μ L of 10 mM AgNO_3 , and 300 μ L of 30 mM sodium citrate were added to transparent 50 mL borosilicate glass vial. Next, 60 μ L of 30% hydrogen peroxide and 250 μ L of 100 mM NaBH_4 aqueous solution were subsequently added. After the 3 h of incubation, the solution color changed to blue and proceed to the growth step without purification.

To the as-synthesized Ag nanoseed solution-contained 50 mL vial, 1 mL of 100 mM L-ascorbic acid and 330 μ L of 75 mM citrate were added. Next, the growth solution was prepared, which composed of 20 mL of 1 mM AgNO_3 and 125 μ L of 100 mM citric acid. The prepared growth solution was added dropwise to the seed solution by the rate of $0.2 \text{ mL} \cdot \text{s}^{-1}$, followed by the color change to deep blue with the additional incubation for 15 min.

Synthesis of IrNPs by hydrothermal galvanic replacement

20 mL of as-synthesized Ag nanoplates and 3 mL of 10 mM IrCl_3 stock solution were homogeneously mixed in a Teflon-lined autoclave vessel. By using the stainless steel autoclave, the mixture was heated for 1 h at 200 °C with gentle magnetic stirring. After the reaction, product was cooled down to room temperature naturally, and the dark-brown colored product was purified by centrifugation for 10 min at 9000 rpm. To remove the unreacted precursors and dissolved ions, product was further washed for 3 times with deionized water. Finally, purified IrNPs were re-dispersed in 5 mL of deionized water for further use.

Characterizations

Morphology of IrNPs was characterized by energy-filtering TEM (LIBRA 120, Carl Zeiss, Germany). HR and HAADF-STEM/EDS mapping were accomplished by a Tecnai F20 (FEI, USA). XPS analysis was performed by photoelectron spectrometer system (K-Alpha+, ThermoFisher Scientific, USA). XRD was performed by SmartLab (Rigaku corporation, Japan). UV-Vis-NIR spectra were obtained by Lambda-465 (PerkinElmer, USA), BioPhotometer D30 spectrophotometer (Eppendorf Co., Hamburg, Germany), and SynergyMx (Biotek, UK). Hydrodynamic diameter and zeta potential were identified by Zetasizer Nano ZS (Malvern, UK). Cell images were obtained by Ti-inverted fluorescence microscope (Nikon Co., Japan) and a CoolSNAP cf charge-coupled device (CCD) camera (Photometrics, USA) with Metamorph image analysis software (Molecular Devices, USA).

POD activity assay of IrNPs

IrNPs POD activity assay was carried out by using TMB, OPDA and DOPA. To the 2 mL of buffered solution with various pH conditions, 40 μL of 10 mM TMB stock solution in DMSO, 20 μL of IrNPs (2 $\mu\text{g Ir/mL}$) were added into quartz cuvette and gently mixed to form homogeneous solution. Take the UV-Vis blank spectrum by as-prepared IrNPs-TMB mixture in buffered solution. POD reaction was started by direct addition of 20 μL of 30 vv% H_2O_2 . UV-Vis spectra were taken in every 10 s.

Bacterial culture and antibacterial experiments

A mono colony of *E. coli* on the solid LB agar plate was transferred to 2 mL of liquid LB culture medium and grown at 37 °C for 18 h under 200 rpm rotation. 100 μL of culture solution were then transferred to 10 mL of liquid LB culture medium and grown for 3.5 h at 37 °C under 200 rpm rotation, until its optical density (OD) became 0.5~0.6. The as-prepared bacteria solution (100 μL) was mixed with varying the concentrations of IrNPs (0.0025, 0.005, 0.01, 0.02 $\mu\text{g Ir/mL}$) and fixed concentration of H_2O_2 (200 μM) for 30 min. Fixed concentration of IrNPs (0.01 $\mu\text{g Ir/mL}$) and varying the concentrations of H_2O_2 (50, 100, 200, 500 μM) mixed with as-prepared bacteria solution in the same manner. Control experiments of bacterial cells only (control), IrNPs (0.01 $\mu\text{g Ir/mL}$), H_2O_2 (200 μM), and dual-treated condition were performed in the same manner. For tracking the growth phase of bacterial cells, OD at 600nm was continuously measured at each 40 min repeatedly by a UV-Vis spectrophotometer.

Cytotoxicity and anticancer effect of IrNPs

HeLa cells were seeded in 96-well culture plate with a density of 10,000 cells/well in total volume of 100 μ L growth media (50-70% confluency). The cells were treated with appropriated concentrations of IrNPs with/without H₂O₂ and incubated for 24 h at 37 °C incubator. After the rinsing with 1xPBS twice, 100 μ L of serum free media with 0.5 mg/mL concentration of MTT solution was added (To prepare MTT stock solution, MTT powder was dissolved in 1xPBS at 5 mg/mL concentration, then filtered through sterilized syringe filter (0.2 mm pore diameter)).

ASSOCIATED CONTENT

Supporting Information. Supporting information contains the additional data including HR-TEM with FFT image, HAADF-STEM (annular bright field and dark field) images, UV-Vis spectra over time.

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